PICKBOT: AUTONOMOUS LOW-COST PICK-AND-PLACE ROBOT

Hackathon Abstract — Proposed System

Problem & Motivation

Industrial robotic arms can automate repetitive material-handling tasks, but their cost and complexity can limit access for small-scale users, educational environments, and early-stage automation projects. PickBot addresses this gap by developing a low-cost robotic platform that can progress from remote operation to autonomous object handling.

Methodology

PickBot is designed as a 5-DOF robotic arm using an ESP32 as the main controller, servo motors for joint movement, and a servo-actuated gripper for object manipulation. A camera-based vision subsystem observes the workspace and provides object information to the control system. The proposed software pipeline performs object detection, selects a target, determines an appropriate pick location, and generates coordinated arm and gripper movements. The ESP32 executes the motion commands while a separate regulated power supply provides suitable power to the actuators. The system is designed with modular mechanical and electronic components so that individual subsystems can be upgraded without redesigning the complete robot.

Proposed Block Diagram

Data flow: visual input → object detection → target selection/motion planning → ESP32 control → servo-driven arm and gripper.

Proposed Outcome

The proposed outcome is a functional autonomous prototype capable of identifying a selected object, moving the robotic arm toward it, gripping it, and placing it at a defined location with minimal human intervention. The project will demonstrate the feasibility of combining affordable embedded hardware, computer vision, and autonomous motion control into a practical pick-and-place platform. Future extensions can include improved object recognition, obstacle avoidance, adaptive gripping, and deployment in small-scale automation and educational applications.

Core Hardware: ESP32 • 5 Servo Motors • Robotic Arm Mechanism • Gripper • Camera/Vision Module • Regulated Power Supply